

Angular Momentum Conservation in the $\text{O} + \text{OH} \leftrightarrow \text{O}_2 + \text{H}$ Reaction

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ABSTRACT: We have studied the $\text{O} + \text{OH} \leftrightarrow \text{O}_2 + \text{H}$ reaction on Varandas's DMBE IV potential using a variety of statistical methods, all involving the RRKM assumption for the HO_2^* complex. Comparing our results using microcanonical variational transition-state theory (μVT) with those using microcanonical/fixed-J variational transition-state theory ($\mu\text{VT-J}$), we find that the effect of angular momentum conservation on the rate coefficient is imperceptible up to a temperature of about 700 K. Above 700 K angular momentum conservation increasingly reduces the rate coefficient, but only by approximately 21% even at 5000 K. Comparing our $\mu\text{VT-J}$ calculations with the quasi-classical trajectory (QCT) results of Miller and Garrett [1], we confirm their conclusion that non-RRKM dynamics of the HO_2^* complex reduces the rate coefficient by about a factor of 2 independent of temperature. Our calculations of $k^{(\text{c})}$, the rate coefficient for HO_2^* formation from $\text{O} + \text{OH}$, are in excellent agreement with the QCT results of Miller and Garrett. Although the differences are not large, we find $k_{\text{CVT}}^{(\text{c})} > k_{\mu\text{VT}}^{(\text{c})} > k_{\mu\text{VT-J}}^{(\text{c})} > k_{\text{QCT}}^{(\text{c})}$, where CVT stands for canonical variational transition-state theory. © 1999 John Wiley & Sons, Inc. *Int J Chem Kinet* 31: 753–756, 1999

INTRODUCTION

Miller and Garrett [1] have studied the title reaction in detail using both quasi-classical trajectory (QCT) and statistical methods. They presented results for both the Varandas DMBE IV [2] and Melius-Blint [3] potentials. The conclusion of their work was that the reaction exhibits marked non-RRKM behavior, the effect of which is to reduce the rate coefficient by approximately a factor of 2 independent of temperature. Their statistical methodology combined the RRKM (or strong coupling) assumption for the HO_2^*

complex with microcanonical variational transition-state theory for the “fluxes” in both the inlet and exit channels. The inability of their statistical methods to conserve angular momentum leaves a lingering doubt that such conservation might also act to reduce their statistical-theoretical rate coefficient significantly. Miller and Garrett argued only qualitatively that angular momentum conservation did not have a significant effect on their results.

In the present short communication we consider the reaction,



and calculate $k^{(\text{c})}$, the rate coefficient for HO_2^* complex formation from $\text{O} + \text{OH}$, and P , the fraction of

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HO₂* complexes so produced that go on to form H + O₂, using statistical methods. For $k^{(c)}$, we use canonical variational transition-state theory (CVT), microcanonical variational transition-state theory (μ VT), and microcanonical/fixed-J variational transition-state theory (μ VT-J) [4]. We compare these results among themselves and with the trajectory results of Miller and Garrett. For P, we use μ VT and μ VT-J, comparing the results with each other, with the trajectory calculations of Miller and Garrett, and with the quantum scattering calculation of Miller [5]. All the theoretical results use Varandas's DMBE IV potential. Making the indicated comparisons on the same potential energy surface (PES) allows us to draw completely unambiguous conclusions about the relative importance of angular momentum conservation and non-RRKM dynamics on the thermal rate coefficient.

THEORY

The O + OH $\rightarrow$ O₂ + H reaction is exothermic by almost 17 kcal/mole and takes place over a potential well approximately 50 kcal/mole deep (measured from H + O₂). Consequently, it would seem reasonable to expect the reaction to obey the RRKM assumption [4], although it apparently does not. Miller and Garrett [1] and Miller [4] have discussed this point in detail, so we shall not belabor it here. Our purpose is simply to calculate, within the context of the RRKM assumption, the effect of conserving angular momentum on the thermal rate coefficient. It should be noted that there is no intrinsic energy barrier separating either the reactants or the products from the HO₂ well.

To accomplish our task it is necessary to define several rate coefficients,

$$k_{\text{CVT}}^{(c)}(T) = \frac{g(T)}{hQ_R(T)} \int_0^\infty N_1(E, \alpha(T)) \times \exp(-E/k_B T) dE \quad (1)$$

$$k_{\mu\text{VT}}^{(c)}(T) = \frac{g(T)}{hQ_R(T)} \int_0^\infty N_1(E, \alpha(E)) \times \exp(-E/k_B T) dE \quad (2)$$

$$k_{\mu\text{VT-J}}^{(c)}(T) = \frac{g(T)}{hQ_R(T)} \sum_J (2J+1) \int_0^\infty N_1(E, J, \alpha(E, J)) \times \exp(-E/k_B T) dE \quad (3)$$

$$k_{\mu\text{VT}}(T) = \frac{g(T)}{hQ_R(T)} \int_0^\infty \frac{N_1(E, \alpha(E))N_2(E, \beta(E))}{N_1(E, \alpha(E)) + N_2(E, \beta(E))} \times \exp(-E/k_B T) dE \quad (4)$$

$$k_{\mu\text{VT-J}}(T) = \frac{g(T)}{hQ_R(T)} \sum_J (2J+1) \times \int_0^\infty \frac{N_1(E, J, \alpha(E, J))N_2(E, J, \beta(E, J))}{N_1(E, J, \alpha(E, J)) + N_2(E, J, \beta(E, J))} \times \exp(-E/k_B T) dE \quad (5)$$

Equations 1–5 are expressions for the overall rate coefficient, k , and that for complex formation, $k^{(c)}$, calculated from canonical variational transition-state theory (CVT), microcanonical variational transition-state theory, and microcanonical/fixed-J variational transition-state theory. In these expressions k_B is Boltzmann's constant, h is Planck's constant, Q_R is the vibrational-rotational-translational partition function of the reactants (not including the contribution from the center-of-mass motion), E is the total energy, J is the total angular momentum quantum number, and $g(T)$ is the ratio of the electronic partition function for the potential on which the reaction takes place to the electronic partition function for the reactants. The $N_i(E, J)$, $i = 1, 2$, are the sums of states on the indicated transition-state dividing surfaces (TSDSs) with energy less than or equal to E and angular momentum quantum number equal to J ; the $N_i(E)$ are defined analogously. The subscripts $i = 1$ and $i = 2$ correspond to the inlet (O + OH) and exit (H + O₂) channels, respectively. The parameters α and β similarly label the TSDSs in the inlet and exit channels. They represent particular choices of TSDS that minimize the appropriate sum of states. There is one TSDS (in each exit channel) for every T in CVT, one for every E in μ VT, and one for every E, J combination in μ VT-J. Note that $N_i(E, J)/h$ and $N_i(E)/h$ can also be interpreted as probability fluxes per unit energy through the i^{th} TSDS.

Given the rate coefficients described above, we can calculate the RRKM probability of HO₂* complexes, formed from O + OH, decomposing into H + O₂ using both microcanonical and microcanonical/fixed-J theories,

$$P_{\mu\text{VT}} = k_{\mu\text{VT}}/k_{\mu\text{VT}}^{(c)} \quad (6)$$

$$\text{and} \quad P_{\mu\text{VT-J}} = k_{\mu\text{VT-J}}/k_{\mu\text{VT-J}}^{(c)} \quad (7)$$

The differences between $P_{\mu\text{VT}}$ and $P_{\mu\text{VT-J}}$ can be attributed to the effects of angular momentum conservation on the total rate coefficient, at least within the context of a statistical RRKM analysis.

The sum of states, either $N_i(E, J)$ or $N_i(E)$, for each channel is calculated from the classical variational principle. The TSDS is chosen from a one-parameter

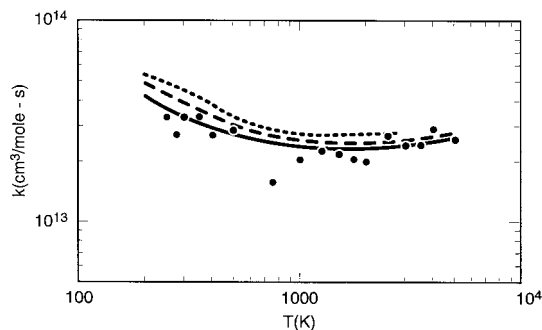


Figure 1 The rate coefficient for complex formation, $k^{(c)}$, as a function of temperature. The points are from the quasi-classical trajectory calculations of Miller and Garrett [1]. The curves are from the present investigation—the short-dashed curve is the canonical variational transition-state theory result ($k_{\text{CVT}}^{(c)}$), the long-dashed curve is from microcanonical variational transition-state theory ($k_{\mu\text{VT}}^{(c)}$) and the solid curve is from microcanonical/fixed-J variational transition-state theory ($k_{\mu\text{VT-J}}^{(c)}$).

family of surfaces orthogonal to the reaction coordinate by minimizing the total flux ($N_i(E, J)/h$ or $N_i(E)/h$) through it. In this investigation we employ methods developed by Klippenstein [6] in calculating the fluxes. These methods are an extension and generalization of the Wardlaw-Marcus approach [7, 8]. The computer code VARIFLEX [9] was used in all our calculations.

RESULTS AND DISCUSSION

Figure 1 summarizes our results for $k^{(c)}$. As expected, $k_{\text{CVT}}^{(c)} > k_{\mu\text{VT}}^{(c)} > k_{\mu\text{VT-J}}^{(c)}$ [4]. However, the differences between the rate coefficients are relatively small — $k_{\text{CVT}}^{(c)}$ is greater than $k_{\mu\text{VT-J}}^{(c)}$ by no more than 35% at any temperature, and $k_{\mu\text{VT}}^{(c)}$ is no greater than 17% larger than $k_{\mu\text{VT-J}}^{(c)}$ anywhere. All three rate coefficients appear to be larger than the QCT results of Miller and Garrett, although the difference between $k_{\text{QCT}}^{(c)}$ and $k_{\mu\text{VT-J}}^{(c)}$ is only barely perceptible. For purely classical calculations, the transition-state theory (TST) rate coefficient would be guaranteed to be larger than the trajectory result by the variational principle [4]. However, quantization of the initial conditions in the QCT approach and the quantum treatment of the “conserved vibration,” together with the assumed decoupling of the conserved and “transitional” modes, in the present implementation of TST compromise this guarantee, at least in principle. Nevertheless, we do find that $k_{\text{CVT}}^{(c)} > k_{\mu\text{VT}}^{(c)} > k_{\mu\text{VT-J}}^{(c)} > k_{\text{QCT}}^{(c)}$, as we would expect

from the variational principle, with the difference between $k_{\text{QCT}}^{(c)}$ and $k_{\mu\text{VT-J}}^{(c)}$ being vanishingly small. Because the methods used here for calculating $k^{(c)}$ are more appropriate for barrierless reactions than those of Miller and Garrett, our TST predictions for $k^{(c)}$ are in substantially better agreement with the trajectory results than are theirs.

Figure 2 shows our results for $P_{\mu\text{VT}}(T)$ and $P_{\mu\text{VT-J}}(T)$, along with a least-squares fit to the trajectory results of Miller and Garrett, $P_{\text{QCT}}(T)$, and the quantum scattering result discussed by Miller [5]. The two statistical probabilities, $P_{\mu\text{VT}}(T)$ and $P_{\mu\text{VT-J}}(T)$, are virtually identical up to a temperature of approximately 700 K. At 2500 K, $P_{\mu\text{VT-J}}$ is smaller than $P_{\mu\text{VT}}$ by 16% and, at 5000 K, by 21%. Therefore, one must conclude that angular momentum conservation has a measurable effect on $k(T)$, at least at sufficiently high temperatures, but that the effect is not a major one. We note that the Miller-Garrett prediction of $P_{\mu\text{VT}}(T)$, although not plotted in Figure 2, is in reasonably good agreement with the present results, lying in between the $P_{\mu\text{VT}}(T)$ and $P_{\mu\text{VT-J}}(T)$ curves of Figure 2.

As discussed by Miller and Garrett, a much more important effect on $k(T)$ is the non-RRKM behavior of the HO_2^* complex. Comparing $P_{\text{QCT}}(T)$ with $P_{\mu\text{VT-J}}(T)$, one finds that this non-RRKM effect acts to reduce the rate coefficient by slightly less than a factor of 2 independent of temperature — $P_{\text{QCT}}/P_{\mu\text{VT-J}} = 0.54$ at 300 K, 0.52 at 2000 K, and 0.53 at 5000 K.

Also shown in Figure 2 is the quantum scattering result of Miller [5] obtained for $J = 0$ at $T = 1200$ K. This calculation is extremely important, because it

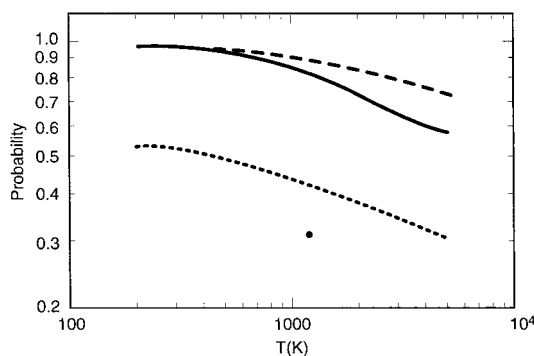


Figure 2 The probability of the HO_2^* complex, formed from $\text{O} + \text{OH}$, dissociating into $\text{H} + \text{O}_2$. The long-dashed curve is the microcanonical prediction, and the solid curve is a least-squares fit to the quasi-classical trajectory calculations of Miller and Garrett [1]. The point is the quantum scattering ($J = 0$) result discussed by Miller [5].

shows that the non-RRKM effect on $k(T)$ persists in the transition from classical to quantum mechanics. Somewhat surprisingly, the effect appears to be larger in quantum mechanics than it is in classical mechanics. However, this may be illusory. Either errors in the (very difficult) quantum calculations or their exclusion of higher J 's may be the cause of the difference between the quantum and QCT scattering predictions of Figure 2. It is not clear what effect including higher J 's in the quantum scattering calculations would have, but the results of such a calculation would complete the comparison.

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